AWS VPC Peering - Vikas

EWB Cloud Labs – AWS VPC Peering (Project 01)

Submission Report

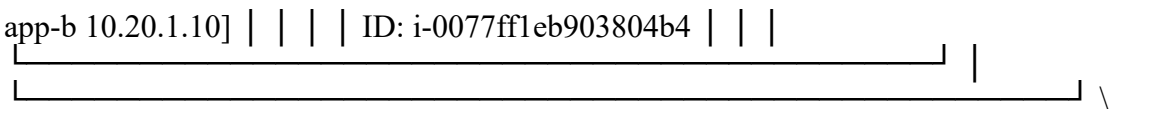
Student: Vikas Marri **Email:** vikasmarri9@gmail.com **Region:** ap-south-1 (Mumbai)
Account ID: 328681353246 **Date:** 2026-10-18

Cover

This report documents the end-to-end build and verification of an AWS VPC Peering connection between two VPCs in the same region and account, following the EWB Cloud Labs Project 01 specification.

Architecture Diagram

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Internet | [ewb-igw-a: igw-0aa7c7acb8d8bdd45] |
┌──┐ │
VPC A: ewb-vpc-a 10.10.0.0/16 │ │ ID: vpc-085f611ada428a582 │ │ │ │
┌──┐ │ │ │ ewb-a-
public 10.10.1.0/24 │ │ │ │ ID: subnet-0504284fed260ff85 │ │ │ │ │ │ │ [ewb-
bastion 10.10.1.10] │ │ │ │ ID: i-08970c0dc4cb29c2b │ │ │ │ Public IP: 15.206.67.80
│ │ │ └──┘ │ │ │
│ ┌──┐ │ │ │ ewb-
a-private 10.10.2.0/24 │ │ │ │ ID: subnet-0dfebabb36e85e00e │ │ │ │ │ │ │ [ewb-
app-a 10.10.2.10] │ │ │ │ ID: i-0ed91a5a28d3d248c │ │ │
└──┘ │
└──┘ │
VPC Peering: pcx-06a06a46ff04e3d8b (ewb-a-b-peer, same account/region) │
┌──┐ │
VPC B: ewb-vpc-b 10.20.0.0/16 │ │ ID: vpc-033d9930b925f06bb │ │ (NO Internet
Gateway) │ │ │ │
┌──┐ │ │ │ ewb-b-
private 10.20.1.0/24 │ │ │ │ ID: subnet-00c65935f6ecc6754 │ │ │ │ │ │ │ [ewb-



Resource Inventory

Resource	ID	CIDR / IP
ewb-vpc-a	vpc-085f611ada428a582	10.10.0.0/16
ewb-a-public	subnet-0504284fed260ff85	10.10.1.0/24
ewb-a-private	subnet-0dfebabb36e85e00e	10.10.2.0/24
ewb-vpc-b	vpc-033d9930b925f06bb	10.20.0.0/16
ewb-b-private	subnet-00c65935f6ecc6754	10.20.1.0/24
ewb-igw-a	igw-0aa7c7acb8d8bdd45	—
ewb-bastion	i-08970c0dc4cb29c2b	10.10.1.10 + 15.206.67.80
ewb-app-a	i-0ed91a5a28d3d248c	10.10.2.10
ewb-app-b	i-0077ff1eb903804b4	10.20.1.10
ewb-rt-a-public	rtb-0c607525197a715e2	—
ewb-rt-a-private	rtb-0e390bcf60f3feabc	—
ewb-rt-b-private	rtb-0fd554d97e9da9953	—
ewb-sg-bastion	sg-021017caf97fe4251	—
ewb-sg-app-a	sg-0d1a7fa830b423a7d	—
ewb-sg-app-b	sg-064ca4b1db6dcf9e3	—
ewb-a-b-peer	pcx-06a06a46ff04e3d8b	—

Route Tables

ewb-rt-a-public (rtb-0c607525197a715e2)

Destination	Target
10.10.0.0/16	local
10.20.0.0/16	pcx-06a06a46ff04e3d8b
0.0.0.0/0	igw-0aa7c7acb8d8bdd45

ewb-rt-a-private (rtb-0e390bcf60f3feabc)

Destination	Target
10.10.0.0/16	local
10.20.0.0/16	pcx-06a06a46ff04e3d8b

ewb-rt-b-private (rtb-0fd554d97e9da9953)

Destination	Target
10.20.0.0/16	local
10.10.0.0/16	pcx-06a06a46ff04e3d8b

Security Group Rules

ewb-sg-bastion (sg-021017caf97fe4251)

Direction	Protocol	Port	Source
Inbound	TCP	22	157.50.x.x/32 (my current public IPv4 only)
Outbound	All	All	0.0.0.0/0

ewb-sg-app-a (sg-0d1a7fa830b423a7d)

Direction	Protocol	Port/Type	Source
Inbound	TCP	22	10.10.1.10/32
Inbound	ICMP	echo-request (8/0)	10.20.1.10/32
Outbound	All	All	0.0.0.0/0

ewb-sg-app-b (sg-064ca4b1db6dcf9e3)

Direction	Protocol	Port/Type	Source
Inbound	TCP	22	10.10.1.10/32
Inbound	ICMP	echo-request (8/0)	10.10.2.10/32
Outbound	All	All	0.0.0.0/0

Peering Status

Connection ID: pcx-06a06a46ff04e3d8b **Requester VPC:** vpc-085f611ada428a582 (10.10.0.0/16) **Accepter VPC:** vpc-033d9930b925f06bb (10.20.0.0/16) **Status:** Active

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Test Outputs

Test 01 – Pre-Peering Failure (ping App-B from App-A)

=== SSH to ewb-bastion: hostname === ip-10-10-1-10.ap-south-1.compute.internal === SSH to ewb-app-a via ProxyJump: hostname === ip-10-10-2-10.ap-south-1.compute.internal === Test 01: ping App-B (10.20.1.10) from App-A - EXPECTED FAILURE === PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.

— 10.20.1.10 ping statistics — 4 packets transmitted, 0 received, 100% packet loss, time 3103ms

exit:1

Test 02 – Forward Ping (App-A → App-B)

=== Test 02: Forward ping App-A -> App-B (10.20.1.10) === PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data. 64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.479 ms 64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.180 ms 64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.172 ms 64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.558 ms

— 10.20.1.10 ping statistics — 4 packets transmitted, 4 received, 0% packet loss, time 3088ms rtt min/avg/max/mdev = 0.172/0.347/0.558/0.173 ms

Test 03 – Reverse Ping (App-B → App-A)

=== Test 03: Reverse ping App-B -> App-A (10.10.2.10) === PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data. 64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.178 ms 64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.259 ms 64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.251 ms 64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.185 ms

— 10.10.2.10 ping statistics — 4 packets transmitted, 4 received, 0% packet loss, time 3111ms rtt min/avg/max/mdev = 0.178/0.218/0.259/0.036 ms

Test 04 – SSH to App-B via ProxyJump

=== Test 04: SSH to App-B (hostname; hostname -I) === ip-10-20-1-10.ap-south-1.compute.internal 10.20.1.10

Test 05 – No Public IP on App-A / App-B

=== Test 05: No public IP on App-A / App-B ===

DescribeInstances |

i-0077ff1eb903804b4 i-0ed91a5a28d3d248c	None None

Controlled Failure and Recovery

=== Step 1: Remove ICMP echo-request rule from ewb-sg-app-b === === Step 2: Ping App-B from App-A (EXPECTED TO FAIL) === PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.

— 10.20.1.10 ping statistics — 4 packets transmitted, 0 received, 100% packet loss, time 3150ms

exit:1 === Step 3: Restore ICMP rule === === Step 4: Ping App-B from App-A (EXPECTED TO SUCCEED after restore) === PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data. 64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.172 ms 64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.183 ms 64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.216 ms 64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.190 ms

— 10.20.1.10 ping statistics — 4 packets transmitted, 4 received, 0% packet loss, time 3120ms rtt min/avg/max/mdev = 0.172/0.190/0.216/0.016 ms

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Forward / Return Path Explanation

Forward path (Bastion SSH → App-A ping → App-B):

1. Packet leaves App-A (10.10.2.10) destined for App-B (10.20.1.10).
2. ewb-rt-a-private matches 10.20.0.0/16 → sends via VPC Peering connection pcx-06a06a46ff04e3d8b.
3. Packet arrives in VPC B; ewb-rt-b-private has no explicit match for 10.10.x.x (only 10.10.0.0/16 → pcx for the return path), local route handles 10.20.0.0/16.
4. ewb-sg-app-b allows ICMP echo-request (type 8) from 10.10.2.10/32.
5. App-B (10.20.1.10) receives the packet and sends echo-reply.

Return path (App-B → App-A):

1. Echo-reply leaves App-B (10.20.1.10) destined for App-A (10.10.2.10).
2. ewb-rt-b-private matches 10.10.0.0/16 → sends via VPC Peering.
3. Packet arrives in VPC A; ewb-rt-a-private matches local 10.10.0.0/16.
4. ewb-sg-app-a allows ICMP echo-request from 10.20.1.10/32; return traffic (echo-reply) is allowed by the stateful security group.
5. App-A receives the reply.

“Be Ready to Explain” Questions

Q1: Why does VPC B have no Internet Gateway? VPC B represents a private backend network. All access is through the peering connection from VPC A. Adding an IGW would expose App-B to the Internet, which violates the least-privilege design.

Q2: Why must the peering connection be explicitly accepted? AWS requires explicit acceptance to prevent one account (or in multi-account setups, one team) from unilaterally accessing another VPC. It is a security control ensuring mutual consent.

Q3: Why did we add a peering route to ewb-rt-a-public? The bastion's ProxyJump to App-B originates from the public subnet. The bastion is the SSH proxy, so its subnet's route table must know how to reach 10.20.0.0/16. Without this route the TCP connection from the bastion to App-B would be dropped.

Q4: Why are the SSH and ICMP security group rules locked to specific /32 CIDRs? 0.0.0.0/0 on SSH or ICMP would allow any host to attempt connections. Using /32 CIDRs (bastion IP or App-A/App-B IPs) limits exposure to only the known hosts that need access, following the principle of least privilege.

What I Learned

1. VPC Peering is non-transitive — traffic between VPC A and VPC C must have its own peering connection; it cannot route through VPC B.
2. Both route tables on both sides of the peering must have matching routes or traffic flows only one way.
3. Security groups are stateful — only inbound rules need to be defined; return traffic is automatically allowed.
4. The ProxyJump technique lets the bastion act as a transparent SSH gateway without copying the private key to the bastion.
5. ICMP rules in AWS use FromPort=type and ToPort=code (e.g., echo-request = 8/0).

Report generated by vpc-peering-lab.ps1 Stage 8

Bastion security group correction

During the build the bastion SSH rule was first set to a wide range (157.50.0.0/16) because my ISP assigns a changing public IP. The workbook requires only my current public IPv4 as a /32, so the rule is now 157.50.x.x/32 and the /16 is removed. SSH to the bastion and ProxyJump to App-A work. Evidence: evidence/02b-bastion-sg-fix.txt. If the ISP changes my IP, the /32 must be updated; widening the rule is not the safe fix.

Console screenshots